Primary 3 · Science

Science · Set 3 — Answer Key

80 marks · 1 h 15 min · Exam mock

For parents. Give this to the pupil only after the paper is finished. Written answers: accept any answer with the same meaning, in a complete sentence with correct tense.

Section A: Multiple-Choice Questions (40 marks)

No.	Answer	Marks
1	(1) Water temperature	2
2	(1) The amount of sugar and number of stirs	2
3	(4) Cup 3	2
4	(1) The temperature of the water	2
5	(1) Seed	2
6	(2) Water and air	2
7	(2) Mushroom	2
8	(4) Mammals	2
9	(3) Lizard	2
10	(4) Some bacteria are helpful, e.g. in making yoghurt	2
11	(4) Cockroach	2
12	(3) Mosquito	2
13	(3) Pupa	2
14	(1) Tail	2
15	(3) Rubber	2
16	(2) An empty plastic bottle	2
17	(4) Clear plastic	2
18	(3) Rainbow	2
19	(2) Fish	2
20	(2) Breaking down dead or decaying matter	2

Section B: Structured Questions (16 marks)

No.	Answer	Marks
21(a)	False	1
21(b)	True	1
22	(1) The surface the car rolls on	2
23	Does not get hot easily (it is a poor conductor of heat, or a good insulator).	2
24	Flexible (elastic) — it can stretch and bend without breaking.	2
25	(3) Fungi cannot make their own food using sunlight	2
26(a)	True	1
26(b)	False	1
27(a)	True	1
27(b)	False	1
28	They grow (or they can reproduce / they respond to their surroundings / they breathe).	2

Section C: Open-Ended Questions (24 marks)

No.	Answer	Marks
29	Only the type of cloth was changed. The size of each cloth, the amount of water, and the time each cloth was dipped in the water were kept the same. This makes it a fair test because any difference in how much water is absorbed is caused only by the type of cloth, not by any other factor. — Marking guide: 1 mark for correctly stating what was changed (the type of cloth), 1 mark for correctly stating what was kept the same (size of cloth, amount of water, dipping time), 1 mark for explaining why this makes the test fair.	3

30	Only the material of the container was changed. The size of the ice cube and the room temperature were kept the same for every container. This makes it a fair test because any difference in how long the ice takes to melt is caused only by the container's material, not by any other factor. — Marking guide: 1 mark for correctly stating what was changed (the material of the container), 1 mark for correctly stating what was kept the same (size of ice cube, room temperature), 1 mark for explaining why this makes the test fair.	3
31	I could gently touch the animal to see if it responds or moves away, and I could watch closely to see if it is breathing, for example by looking for small movements near its breathing holes. — Marking guide: 1 mark for a sensible test of response to touch/stimulus, 1 mark for a second sensible test (e.g. checking for breathing or other movement), 1 mark for a clear, complete explanation. Give 0 marks for actions that are CARE rather than a TEST (e.g. giving it food or water, or moving it somewhere safer) — these do not test whether it is alive.	3
32	The mushrooms are fungi, so they cannot make their own food using sunlight. Instead, they feed on the dead wood of the log, breaking it down bit by bit. Over time, this causes the log to rot away and break into smaller pieces. — Marking guide: 1 mark for stating that fungi feed on the dead/decaying wood (cannot make their own food), 1 mark for describing that this breaks the wood down, 1 mark for the missing link — that this is why the log eventually rots away / disappears over time.	3
33	A cockroach has no pupa stage — its nymph already looks similar to the adult and simply grows bigger after each moult. A beetle, however, goes through a pupa stage, during which its body changes completely from a larva into an adult, which the cockroach never goes through. — Marking guide: 1 mark for describing the cockroach nymph as already resembling the adult with no big body change, 1 mark for describing the beetle's pupa stage as a stage of major body change, 1 mark for a clear comparison between the two.	3
34	Plastic is a good choice because it is waterproof, so rain does not soak through and wet the person wearing it. It is also flexible, so it is comfortable to move around in and does not crack easily. — Marking guide: 1 mark for stating plastic is waterproof, 1 mark for a second correct property (e.g. flexible, lightweight), 1 mark for a clear overall explanation linking the properties to why plastic suits a raincoat.	3

35	Although both mushrooms and moss can be found growing in damp, shady places, the key difference is that moss, like other plants, can make its own food using sunlight, but mushrooms, which are fungi, cannot make their own food — they get their food by feeding on dead or decaying matter instead. — Marking guide: 2 marks for stating the key difference clearly for BOTH sides — fungi like mushrooms cannot make their own food, but plants like moss can make their own food using sunlight (1 mark for each side). 1 mark for noting that growing in the same kind of place does not make them the same kind of living thing.	3
36	A seed needs water (and air) to germinate and start growing. Seed A had water, so it was able to germinate and sprout. Seed B had no water, so it could not start growing even though it is a living seed, and it stayed the same. — Marking guide: 1 mark for stating that a seed needs water to germinate, 1 mark for linking this to Seed A sprouting because it had water, 1 mark for the missing link — explaining that Seed B did not germinate because it had no water, even though it is alive.	3